5. COMPUTING OPTIMAL EXPONENT PAIRS

5.1 INTRODUCTION

In the last chapter, we saw several applications of exponent pairs where the ultimate result was expressed as a function of

$$\theta(k,l) = \frac{ak + bl + c}{dk + el + f} \tag{5.1.1}$$

where $a, b, \ldots, f$ are real numbers. In this chapter, we give an algorithm for computing $\inf \theta$, where the infimum is taken over all exponent pairs produced by the A- and B-processes.

Let $\mathbf{P}$ denote the set of all exponent pairs produced from $(0,1)$ by the A- and B-processes. Since B^2 is the identity, we may write $\mathbf{P} = A\mathbf{P} \cup BA\mathbf{P}$, where

$$A\mathbf{P} = \{A(k,l) : (k,l) \in \mathbf{P}\} \text{ and } BA\mathbf{P} = \{BA(k,l) : (k,l) \in \mathbf{P}\}.$$

Note that $(0,1) \in A\mathbf{P}$ since $A(0,1) = (0,1)$. The other exponent pairs in $A\mathbf{P}$ have the form

$$A^{q_1}BA^{q_2}B \ldots A^{q_k}B(0,1).$$

Similarly, the exponent pairs in $BA\mathbf{P}$ have the form

$$BA^{q_1}BA^{q_2}B \ldots A^{q_k}B(0,1).$$

The A and B processes may be regarded as linear transformations on projective space. Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 & -1 \\ 2 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

Then

$$A \begin{bmatrix} k \\ l \\ 1 \end{bmatrix} = \begin{bmatrix} k \\ k+l+1 \\ 2k+2 \end{bmatrix}$$

and in projective space, this is equivalent to

$$\begin{bmatrix} k/(2k+2) \\ (k+l+1)/(2k+2) \\ 1 \end{bmatrix} = \begin{bmatrix} \kappa \\ \lambda \\ 1 \end{bmatrix},$$

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where $(\kappa, \lambda) = A(k, l)$. A similar relation holds for the B matrix. We are of course, abusing notation by using the letters A and B in two different senses, but the precise meaning should be clear from context. The matrix representations are particularly useful for computing compositions. For example, the matrix representing the composition $A(B(k, l))$ is simply AB.

Let θ be as defined in (5.1.1). The notation $\inf \theta$ will be understood to mean

$$\inf\{\theta(k, l) : (k, l) \in \mathbf{P}\}.$$

Similarly, if $\mathbf{Q} \subseteq \mathbf{P}$, then

$$\inf_{\mathbf{Q}} \theta = \inf\{\theta(k, l) : (k, l) \in \mathbf{Q}\}.$$

By Lemma 2.3, exponent pairs have a convexity property; that is, if (k_1, l_1) and (k_2, l_2) are exponent pairs then so is $(tk_1 + (1-t)k_2, tl_1 + (1-t)l_2)$ for any t with $0 \le t \le 1$. Thus the convex hull of $\mathbf{P}$, which we will denote by conv $\mathbf{P}$, is a set of exponent pairs.

Throughout most of this chapter, we will be working with $\mathbf{P}$ instead of conv $\mathbf{P}$. One reason for this is that for the functions θ that are considered here,

$$\inf_{\mathbf{P}} \theta = \inf_{\text{conv } \mathbf{P}} \theta$$

(see Lemma 5.4). Moreover, we could state an analog of Lemma 5.1 for conv $\mathbf{P}$, but it would be more complicated. On the other hand, certain arguments can be made simpler by appropriate use of convexity; two such arguments are Applications 3 and 4 of Section 5.4.

5.2 PRELIMINARY LEMMAS

Our first lemma collects together several simple facts about the geometry of $\mathbf{P}$.

LEMMA 5.1.

(a) *If* $(k, l) \in \mathbf{P}$ *then* $k + l \le 1$.

(b) *If* $(k, l) \in A\mathbf{P}$ *then* $l - k - 1/2 \ge 0$.

(c) *If* $(k, l) \in BA\mathbf{P}$ *then* $l - k - 1/2 \le 0$.

(d) $A\mathbf{P} \cap BA\mathbf{P} = \{(1/6, 2/3)\}$.

Proof. If $(k, l) \in \mathbf{P}$ then either $(k, l) = A(\kappa, \lambda)$ or $(k, l) = BA(\kappa, \lambda)$ for some $(\kappa, \lambda) \in \mathbf{P}$. In either case,

$$k + l = \frac{2\kappa + \lambda + 1}{2\kappa + 2} \le 1.$$

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This proves (a). To prove (b), suppose $(k,l) = A(\kappa,\lambda)$. Then

$$l - k - 1/2 = \frac{\lambda - \kappa}{2\kappa + 2} \geq 0 \tag{5.2.1}$$

since $\lambda \geq 1/2 \geq \kappa$. Part (c) follows from (a) and (b) and the observation that B is a reflection through the line $l - k - 1/2$. Part (d) follows from (b),(c), and the observation that equality holds in (5.2.1) if and only if $(\kappa,\lambda) = (1/2,1/2)$. Note that $(1/6,2/3) = AB(0,1) = BAB(0,1)$.

In geometric terms, (a) states that $\mathbf{P}$ lies in the triangle with vertices $(0,1)$, $(0,1/2)$, and $(1/2,1/2)$. In fact, it is easy to show that for any $q \geq 0$, $A^q\mathbf{P}$ lies in the triangle with vertices $(0,1)$, $(0,1/2)$, and $A^q(1/2,1/2)$.

We now give three lemmas on θ. We shall need to assume that

$$dk + el + f > 0 \tag{5.2.2}$$

for $(k,l) \in \mathbf{P}$. In practice, it is usually easier to establish that (5.2.2) holds for $(k,l) = (0,1/2),(0,1)$, and $(1/2,1/2)$. This is sufficient, for if (5.2.2) holds for these three points, it must also hold in the convex hull of these points.

We may regard θ as a 2×3 matrix, that is,

$$\theta = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}.$$

We denote the three subdeterminants of θ by

$$u = bf - ce, v = af - cd, \text{ and } w = ae - bd. \tag{5.2.3}$$

Furthermore, let

$$\xi(\theta) = \begin{bmatrix} u \\ v \\ w \end{bmatrix}.$$

We shall see later that computing $\inf \theta$ depends only on the projective equivalence class of $\xi(\theta)$.

LEMMA 5.2. *Suppose that θ is defined by (5.1.1) and let u,v, and w be defined by (5.2.3). Then for any ordered pairs (k,l) and (κ,λ),*

$$(\theta(k,l) - \theta(\kappa,\lambda))(dk + el + f)(d\kappa + e\lambda + f) = \det \begin{bmatrix} k & l & 1 \\ \kappa & \lambda & 1 \\ u & -v & w \end{bmatrix}.$$

The proof is a straightforward computation, and it is left to the reader. This lemma is actually a special case of the Cauchy-Binet formula.

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LEMMA 5.3. *Let θ, u, v, and w be as in the previous lemma. Suppose also that θ satisfies (5.2.2). If $(k,l) \in A\mathbf{P}$ and $(k,l) \neq (1/6, 2/3)$ then $\theta B(k,l) - \theta(k,l)$ has the same sign as $w(k+l) + v - u$.*

Proof. By Lemma 5.2 and (5.2.2), $\theta B(k,l) - \theta(k,l)$ has the same sign as

$$\det \begin{bmatrix} l - 1/2 & k + 1/2 & 1 \\ k & l & 1 \\ u & -v & w \end{bmatrix}.$$

A few calculations with row operations show that this is equal to $(l - k - 1/2) \times (w(k+l) + v - u)$. The first factor does not affect the sign since it is positive by parts (b), (c), and (d) of Lemma 5.1.

LEMMA 5.4. *Let θ be as in Lemma 5.2. Assume that (k_0, l_0) and (k_1, l_1) are exponent pairs and that*

$$\theta(k_0, l_0) \leq \theta(k_1, l_1).$$

For t with $0 \leq t \leq 1$, define

$$(k_t, l_t) = (1 - t)(k_0, l_0) + t(k_1, l_1).$$

Then

$$\theta(k_0, l_0) \leq \theta(k_t, l_t) \leq \theta(k_1, l_1).$$

Proof. From Lemma 5.2, we see that $\theta(k_t, l_t) - \theta(k_0, l_0)$ has the same sign as

$$\det \begin{bmatrix} k_t & l_t & 1 \\ k_0 & l_0 & 1 \\ u & -v & w \end{bmatrix} = t \det \begin{bmatrix} k_1 & l_1 & 1 \\ k_0 & l_0 & 1 \\ u & -v & w \end{bmatrix}.$$

The right-hand side is non-negative by hypothesis, so $\theta(k_t, l_t) \geq \theta(k_0, l_0)$. A similar argument shows that $\theta(k_1, l_1) \geq \theta(k_t, l_t)$.

5.3 THE ALGORITHM

Since $\mathbf{P} = A\mathbf{P} \cup BA\mathbf{P}$, it follows that either $\inf \theta = \inf \theta A$ or $\inf \theta = \inf \theta BA$. Our next theorem decides which option is true in most circumstances.

THEOREM 5.5. *Let θ, defined by (5.1.1), be a function satisfying (5.2.2). Let u, v, and w be defined by (5.2.3). Let r be any real number such that*

$$r \leq \inf (k + l).$$

Define

$$Y = \max(wr + v - u, w + v - u)$$

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and

$$Z = \min(wr + v - u, w + v - u).$$

If $Z \geq 0$, *then* $\inf \theta = \inf \theta A$. *If* $Y \leq 0$, *then* $\inf \theta = \inf \theta BA$.

Note that Y and Z depend only on $\xi(\theta)$. Moreover, if $\xi(\theta)$ is multiplied by a positive constant, the signs of Y and Z do not change. This justifies our earlier remark that $\inf \theta$ depends only on the projective equivalence class of θ.

Theorem 5.5 gives no information when $Z < 0 < Y$. We shall refer to this as the *indeterminate case.* In this instance, we resort to Theorem 5.6.

Proof. If $(k, l) \in A\mathbf{P}$, then $Z \leq w(k + l) + v - u \leq Y$. Assume that $Z \geq 0$. By Lemma 5.3, we see that $\theta B(k, l) \geq \theta(k, l)$ for $(k, l) \in A\mathbf{P}$. Consequently,

$$\inf_{BA\mathbf{P}} \theta \geq \inf_{A\mathbf{P}} \theta.$$

Since $\mathbf{P} = A\mathbf{P} \cup BA\mathbf{P}$, we see that $\inf \theta = \inf \theta A$.

On the other hand, if $Y \leq 0$ then $\theta B(k, l) \leq \theta(k, l)$ for all $(k, l) \in A\mathbf{P}$, and so $\inf \theta = \inf \theta BA$.

To apply Theorem 5.5, it is necessary to have a good lower bound for $\inf k + l$. The lower bound $r = 1/2$ follows from the definition of exponent pairs. We can use Theorem 5.5 with this value of r and

$$\theta = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We find that $Y = Z = 0$, so $\inf \theta = \inf \theta A = \inf \theta BA$. We arbitrarily choose θA, and we set

$$\theta_1 = \theta A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix}.$$

Since $\xi(\theta_1) = [1 \quad 1 \quad -1]^T$, we get $Y = -1/2$ and $\inf \theta = \inf \theta ABA$. Now

$$\theta ABA = \begin{bmatrix} 4 & 2 & 3 \\ 4 & 2 & 4 \end{bmatrix}.$$

On the triangle with vertices $(0, 1), (0, 1/2)$, and $(1/2, 1/2)$, the function $(4k + 2l + 3)/(4k + 2l + 4)$ assumes its minimum at $(0, 1/2)$. Therefore $\inf k + l \geq 4/5$.

Continuing in this fashion, we find that

$$\begin{aligned} \inf k + l \geq & \theta ABA^3BA^2BABABA^2BABABA^2BA^2BABA^2BA^2BA^2BABA(0, 1/2) \\ = & \frac{89037450}{107400671} > 0.8290213568591. \end{aligned}$$

The next result provides a means of analyzing the indeterminate case. It does not provide an answer for every possible θ in the indeterminate case, but we have found it to be sufficient for our applications.

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THEOREM 5.6. *Let θ and r be as in Theorem 5.5. Let C be some finite product of A's and B's such that*

$$\inf \theta BA = \inf \theta BAC.$$

Suppose further that

$$\sup\{k + l : (k, l) \in CA\mathbf{P}\} = r_1.$$

If $\min(rw + v - u, r_1 w + v - u) \geq 0$, *then* $\inf \theta = \inf \theta A$.

Proof. Assume $(k, l) \in BAC\mathbf{P}$. By hypothesis, $k + l \leq r_1$. This together with our other hypotheses and Lemma 5.3 yields $\theta B(k, l) \geq \theta(k, l)$. Consequently,

$$\inf_{\mathbf{P}} \theta \leq \inf_{AC\mathbf{P}} \theta \leq \inf_{AC\mathbf{P}} \theta B = \inf_{\mathbf{P}} \theta BAC = \inf_{\mathbf{P}} \theta BA.$$

To illustrate how Theorem 5.6 can be used, consider the following example, which arises in the computation of $\inf k + l$. Suppose θ is such that

$$\xi(\theta) = [u \quad v \quad w]^T = [386 \quad 801 \quad -450]^T.$$

Now compute Z and Y from the formulas given Theorem 5.5. We find that $Z < 0 < Y$, so θ is in the indeterminate case. Now consider θBA. Note that

$$\xi(\theta BA) = [576 \quad 386 \quad -351]^T;$$

consequently, $\inf \theta BA = \inf \theta BABA$. Now we claim that if $(k, l) \in ABA\mathbf{P}$ then $k + l \leq 6/7$. For if $(k, l) = ABA(\kappa, \lambda)$, then $k + l = 1 - 1/(4\kappa + 2\lambda + 4)$, and $4\kappa + 2\lambda + 4 \leq 7$ by Lemma 5.1(a). Going back to $\xi(\theta)$, we find that

$$\min(rw + v - u, 6w/7 + v - u) \geq 0,$$

so $\inf \theta = \inf \theta A$.

In the next corollary, we collect three special cases of Theorem 5.6. We omit the proofs since they are very similar to the discussion in the previous paragraph.

COROLLARY 5.7. *Let θ and r be as in Theorem 5.5. If any of the following hypotheses hold, then* $\inf \theta = \inf \theta A$.

(a) $\inf \theta BA = \inf \theta BABA$ and $\min(rw + v - u, 6w/7 + v - u) \geq 0$.

(b) $\inf \theta BA = \inf \theta BABA^2$ and $\min(rw + v - u, 5w/6 + v - u) \geq 0$.

(c) $\inf \theta BA = \inf \theta BABA^3BA$ and $\min(rw + v - u, 73w/88 + v - u) \geq 0$.

The main part of our algorithm consists of examining $\xi(\theta)$ and using Theorem 5.6 or Corollary 5.7 to determine whether $\inf \theta = \inf \theta A$ or $\inf \theta = \inf \theta BA$. We then replace $\xi(\theta)$ by the appropriate choice of $\xi(\theta A)$ or $\xi(\theta BA)$, and we repeat the process. After an appropriate number of iterations, we expect to have

$$\inf \theta = \inf \theta B^j A^{q_1} BA^{q_2} B \dots A^{q_r} BA,$$

where $j = 0$ or 1 and $q_1, q_2, \dots, q_r$ are positive integers. However, it is possible to have

$$\inf \theta = \inf \theta A^q$$

for every $q \geq 0$. The next theorem gives us an easy way of recognizing when this happens.

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THEOREM 5.8. *Let θ, u, v, and w be as in Theorem 5.5. The following conditions are equivalent.*

(a) $\inf \theta = \inf \theta A^q B$ for every $q \geq 0$.

(b) $\inf \theta = \theta(0,1)$.

(c) $w + v \geq u$ and $u \leq 0$.

Proof. We shall first prove that (a) and (b) are equivalent, and then we shall complete the proof by showing that (b) and (c) are equivalent.

Assume first that $\inf \theta = \inf \theta A^q$ for every $q \geq 0$. Then we can find a sequence $\{(k_q, l_q)\}$ of exponent pairs such that $(k_q, l_q) \in A^q \mathbf{P}$ and

$$\inf \theta = \lim_{q \to \infty} \theta(k_q, l_q).$$

Now we can show by induction that

$$A^q = \begin{bmatrix} 1 & 0 & 0 \\ Q-q-1 & 1 & Q-1 \\ 2Q-2 & 0 & Q \end{bmatrix}, \tag{5.3.1}$$

where $Q = 2^q$. Therefore, if $(k_q, l_q) = A(k, l)$ then

$$(k_q, l_q) = \left(\frac{k}{(2Q-2)k + Q}, 1 - \frac{k + l - 1}{(2Q-2)k + Q}\right).$$

It follows that $k_q \leq 1/Q$ and $1 - l_q \leq (q+1)/2Q$; therefore

$$\lim_{q \to \infty} (k_q, l_q) = (0, 1).$$

Now θ is uniformly continuous on the convex hull of the triangle with vertices $(0, 1/2)$, $(0,1)$, and $(1/2, 1/2)$, so

$$\lim_{q \to \infty} \theta(k_q, l_q) = \theta(0,1).$$

Now assume that $\inf \theta = \theta(0,1)$. For any $q \geq 0$, $A^q \mathbf{P} \subseteq \mathbf{P}$, so $\inf \theta \leq \inf \theta A^q$. On the other hand, $(0,1) = A^q(0,1)$, so

$$\inf \theta A^q \leq \theta(0,1) = \inf \theta.$$

This completes the proof that (a) and (b) are equivalent.

Next, we prove that (b) and (c) are equivalent. By Lemma 5.2, $\theta(0,1) - \theta(k,l)$ has the same sign as

$$T = u(1 - l - k) - k(w + v - u).$$

If $w + v - u \geq 0$ and $u \leq 0$, then $T \leq 0$ for all $(k,l) \in \mathbf{P}$, so $\inf \theta = \theta(0,1)$. Conversely, assume that $\inf \theta = \theta(0,1)$. Then $T \leq 0$ for all $(k,l) \in \mathbf{P}$. In particular, $T \leq 0$ for $(k,l) = A^q B(0,1)$. From (5.3.1) we see that

$$A^q B(0,1) = \left(\frac{1}{4Q-2}, 1 - \frac{q+1}{4Q-2}\right),$$

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where $Q = 2^q$. Therefore

$$qu \leq w + v - u \tag{5.3.2}$$

for all non-negative integers q. From this with $q = 0$, we see that $w + v - u \geq 0$. We also see that $u \leq 0$, for otherwise (5.3.2) would be false for q sufficiently large. This completes the proof.

The example $\theta(k,l) = k$ shows that it is possible to have $\inf \theta = \theta(0,1)$. The example $\theta(k,l) = l$ shows that it is possible to have $\inf \theta = \theta(\frac{1}{2}, \frac{1}{2})$. The following theorem shows that this property holds for no other element of $\mathbf{P}$.

THEOREM 5.9. *Suppose θ satisfies the conditions of Theorem 5.5, and suppose there is some $(k,l) \in \mathbf{P}$ such that $\inf \theta = \theta(k,l)$. Then either $\inf \theta = \theta(0,1)$ or $\inf \theta = \theta(1/2, 1/2)$.*

Proof. Suppose there is some $(k,l) \in \mathbf{P}$ such that $\inf \theta = \theta(k,l)$. Now (k,l) must have the form

$$B^j A^{q_1} BA \ldots A^{q_r} B(0,1),$$

where $j = 0$ or 1 and the q_i's are positive integers. If $r = 0$ then $(k,l) = (1/2, 1/2)$ or $(0,1)$, and we are done. Assume that $r \geq 1$, and let

$$\theta_1 = \theta B^j A^{q_1} BA \ldots A^{q_r - 1} BA^{q_r - 1}.$$

Now $\inf \theta \leq \inf \theta_1$ since

$$B^j A^{q_1} BA \ldots A^{q_r - 1} BA^{q_r - 1} \mathbf{P} \subseteq \mathbf{P}.$$

Furthermore, $\inf \theta = \theta_1 AB(0,1) \geq \inf \theta_1$. Therefore, $\inf \theta_1 = \theta_1(\frac{1}{6}, \frac{2}{3})$.

Now $ABA^3B(0,1) = (\frac{11}{82}, \frac{57}{82})$ and $BABA^3B(0,1) = (\frac{8}{41}, \frac{26}{41})$. Consequently,

$$\theta_1(\frac{1}{6}, \frac{2}{3}) \leq \min\{\theta_1(\frac{11}{82}, \frac{57}{82}), \theta_1(\frac{8}{41}, \frac{26}{41})\} \leq \theta_1(\frac{27}{164}, \frac{109}{164}); \tag{5.3.3}$$

the last inequality following by an application of Lemma 5.4. In a similar fashion, we also obtain

$$\theta_1(\frac{1}{6}, \frac{2}{3}) \leq \min\{\theta_1(0,1), \theta_1(\frac{1}{2}, \frac{1}{2})\} \leq \theta_1(\frac{1}{4}, \frac{3}{4}). \tag{5.3.4}$$

Now $(\frac{1}{6}, \frac{2}{3})$ lies on the line segment joining $(\frac{27}{164}, \frac{109}{164})$ and $(\frac{1}{4}, \frac{3}{4})$. Thus we may apply Lemma 5.4 again to get

$$\theta_1(\frac{1}{6}, \frac{2}{3}) \geq \min\{\theta_1(\frac{27}{164}, \frac{109}{164}), \theta_1(\frac{1}{4}, \frac{3}{4})\}. \tag{5.3.5}$$

From (5.3.3) through (5.3.5), we see that

$$\theta_1(\frac{1}{6}, \frac{2}{3}) = \min\{\theta_1(\frac{27}{164}, \frac{109}{164}), \theta_1(\frac{1}{4}, \frac{3}{4})\}.$$

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From the proof of Lemma 5.4, we see that this can happen only if

$$\theta_1(\frac{27}{164},\frac{109}{164}) = \theta_1(\frac{1}{6},\frac{2}{3}) = \theta_1(\frac{1}{4},\frac{3}{4}).$$

This together with (5.3.4) shows that

$$\theta_1(\frac{1}{4},\frac{3}{4}) = \min\{\theta_1(0,1), \theta_1(\frac{1}{2},\frac{1}{2})\}.$$

Using Lemma 5.4 again, we get

$$\theta_1(\frac{1}{6},\frac{2}{3}) = \theta_1(\frac{1}{4},\frac{3}{4}) = \theta(0,1) = \theta(\frac{1}{2},\frac{1}{2});$$

therefore $\inf\theta_1 = \theta_1(0,1)$. Now

$$\theta_1(0,1) = \theta B^j A^{q_1} B \dots A^{q_{r-1}} B A^{q_r - 1}(0,1) = \theta B^j A^{q_1} B \dots A^{q_{r-1}} B(0,1).$$

In other words, $\inf\theta = \theta(k_1, l_1)$, where

$$(k_1, l_1) = B^j A^{q_1} B \dots A^{q_{r-1}} B(0,1).$$

Repeating this argument, we get $\inf\theta = \inf\theta(k_r, l_r)$, where $(k_r, l_r) = B^j(0,1)$ is either $(0,1)$ or $(1/2, 1/2)$. This completes the proof.

We can now give a step-by-step description of an algorithm for determining optimal exponent pairs.

1. Check θ for condition (5.2.2); that is, $dk + el + f \geq 0$.
2. Compute $\xi(\theta)$.
3. Use Theorem 5.8 to see if $\inf\theta = \theta(0,1)$. If so, then stop.
4. Use Theorem 5.8 on θB to see if $\inf\theta = \theta(\frac{1}{2},\frac{1}{2})$. If so, then stop.
5. Use Theorem 5.5 to see if $\inf\theta = \theta A$ or $\inf\theta = \theta BA$. If this fails, try Theorem 5.6. If this fails, give up.
6. If $\inf\theta = \inf\theta A$, replace $\xi(\theta)$ by $\xi(\theta A)$. If $\inf\theta = \inf\theta BA$, replace $\xi(\theta)$ by $\xi(\theta BA)$. In either case, return to step 5.

Theorem 5.9 shows that is unnecessary to repeat steps 2 and 3 after the first loop. It is also unnecessary to repeat step 1 after the first loop. To see why, suppose that θ satisfies (5.2.2) and that $\inf\theta = \inf\theta A$, for example. Let

$$\theta A = \begin{bmatrix} a_1 & b_1 & c_1 \\ d_1 & e_1 & f_1 \end{bmatrix}.$$

Then

$$d_1 k + e_1 l + f = d\kappa + e\lambda + f,$$

where $(k,l) = A(\kappa,\lambda)$. The right hand side is positive since (5.2.2) is satisfied for θ.

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5.4 APPLICATIONS

The results of the algorithm of the preceding section may be regarded as a sequence of exponent pairs that yield approximations to the desired infimum. The terms in this sequence are

$$A^{q_1}B(0,1)\,,\ A^{q_1}BA^{q_2}B(0,1)\,,$$

et cetera. We shall call the sequence of exponents $(q_1, q_2, q_3, \ldots)$ the q-sequence. Furthermore, we shall use score-by-innings notation and write the q-sequence as

$$q_1 q_2 q_3 \cdots$$

In this section, we will give the results that algorithm yields when applied to the problems discussed in Chapter 4. The computations reported here were done with MACSYMA.

Rankin's Constant. The quantity

$$\inf k + l$$

plays an important role in our algorithm. We call this quantity Rankin's Constant, and we denote it by the letter R. The first 50 terms in the q-sequence are

13211 21122 12221 21122 11213 21112 11132 11132 11221 11122.

From these terms, we find that

$$R = 0.829021356859133592409239777283112050988343270 3 \pm 8 \times 10^{-43}.$$

In some of the other applications discussed in Chapter 4, the optimal results can be expressed in terms of R. For example, let

$$\theta(k,l) = \frac{k+l}{2k+2}.$$

and let $\alpha = \inf \theta$. We see from Theorems 4.6 and 4.8 that

$$\Delta(x) \ll x^{\alpha+\epsilon} \text{ and } R(x) \ll x^{\alpha+\epsilon}$$

for every $\epsilon > 0$. Let $\theta_1(k,l) = k + l$. As we mentioned in the previous section, $\inf \theta_1 = \inf \theta_1 A$. Moreover,

$$\theta_1 A(k,l) = \frac{2k+l+1}{2k+2} = \theta(k,l) + \frac{1}{2}.$$

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Therefore $\alpha = R - 1/2 = 0.32902135685913359240923977 72\ldots$.

In Theorem 4.13, we showed that if

$$\theta(k,l) = \frac{4k+l+3}{4k+4}$$

and $c < (\theta(k,l))^{-1}$, then

$$\pi_c \sim \frac{x}{c \log x}. \tag{5.4.1}$$

Again, let $\theta_1(k,l) = k + l$. Since

$$\theta(k,l) = \frac{1}{2}(\theta_1 A + 1)$$

we see that $\inf \theta = (R+1)/2$. Therefore, (5.4.1) is true for

$$c < (\inf \theta)^{-1} = 1.0934809440576886234567678933665528741229168\ldots.$$

Heath-Brown's improvement of Piatetski-Shapiro. Lest the reader obtain the impression that all applications of the algorithm reduce to computing R, we mention the following result due to Heath-Brown (Heath-Brown (1983)). If (k,l) is an exponent pair and

$$\theta(k,l) = \frac{12k+10}{12k-2l+13},$$

then $\pi_c \sim x/(c \log x)$ for $c \leq (\theta(k,l))^{-1}$. The first 50 terms of the q-sequence are

01222 11111 11121 11211 11121 12121 12123 21211 12211 11111.

It follows that (5.4.1) is true for

$$c < 1.1404913648148855613299352226496085 54\ldots.$$

Gaps between square-free numbers. Let

$$S(N) = \sum_{N < n \leq 2N} \psi(xn^{-2}).$$

From Lemma 4.3, we see that if (k,l) is an exponent pair and $N = x^{\alpha}$ then

$$S(N) \ll x^{\theta_\alpha(k,l)},$$

where

$$\theta_\alpha(k,l) = \frac{k + (l-2k)\alpha}{k+1}$$

For fixed α, let $g(\alpha) = \inf \theta_\alpha(k,l)$. From Lemma 4.9, we see that if

$$\delta > \sup\{g(\alpha) : 0 \leq \alpha \leq 1/2\},$$

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then there is a square-free number n in the interval

$$x - x^{\delta} < n \leq x$$

when x is sufficiently large.

We can use the algorithm of Section 5.3 to compute $g(\alpha)$ for any fixed α, but that algorithm gives us no direct method of computing $\sup g(\alpha)$. Instead, we use the following argument to show that

$$\sup g(\alpha) = \frac{2R+1}{6R+7}.$$

We begin by treating the range $\alpha < 3/10$. Let

$$(k,l) = BABABA^2BAB(0,1) = (\frac{13}{49}, \frac{57}{98}).$$

Since $l \geq 2k$, we see that

$$\theta\alpha(k,l) \leq \theta_{\frac{3}{10}}(k,l) = \frac{55}{248} < \frac{(2R+1)}{(6R+7)}$$

when $\alpha \leq 3/10$.

Now suppose $\alpha \geq 4/13$. If

$$(k,l) = BA^2BA^2BAB(0,1) = (\frac{19}{62}, \frac{52}{93})$$

then $l > 2k$, and so

$$\theta_{\alpha}(k,l) \leq \theta_{\frac{4}{13}}(k,l) = \frac{701}{3159} < \frac{(2R+1)}{(6R+7)}.$$

We may therefore assume henceforth that $3/10 < \alpha < 4/13$.

We can use Theorem 5.5 to show that $\inf \theta_{\alpha} = \inf \theta_{\alpha} BA$ when $3/10 < \alpha < 4/13$. We therefore set

$$\psi_{\alpha}(k,l) = \theta_{\alpha} BA(k,l) = \frac{l + \alpha(2k - 2l + 1)}{2k + l + 2},$$

so that $g(\alpha) = \inf \psi_{\alpha}$.

We next show that $g(\alpha) \leq (2R+1)/(6R+7)$ for all α under consideration. For (k,l) in $\mathbf{P}$, use convexity to construct the exponent pair

$$(k',l') = \frac{1}{2}((k,l) + B(k,l)) = \frac{1}{4}(2k + 2l - 1, 2k + 2l + 1).$$

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By Lemma 5.4,

$$g(\alpha) \le \inf \psi_\alpha(k', l') = \inf \frac{2(k+l)+1}{6(k+l)+7} = \frac{2R+1}{6R+7}.$$

We now show that there is some β such that $g(\beta) \ge (2R+1)/(6R+7)$. Define $g_A = \inf \psi_\alpha A$ and $g_B = \inf \psi_\alpha BA$, so that $g(\alpha) = \min(g_A(\alpha), g_B(\alpha))$. From Lemma 5.1(b) we see that for fixed (k, l), $\psi_\alpha A(k, l)$ is an decreasing function of α. Therefore $g_A(\alpha)$ is decreasing. Similarly, $g_B(\alpha)$ is an increasing function. From Theorem 5.5, we see that $g_B(3/10) < g_A(3/10)$ and $g_B(4/13) > g_A(4/13)$. Now g_A and g_B are continuous, so there is some β for which $g_A(\beta) = g_B(\beta)$.

Now let $\{(k'_n, l'_n)\}$ and $\{(k''_n, l''_n)\}$ be sequences of exponent pairs (in $A\mathbf{P}$ and $BA\mathbf{P}$, respectively) such that

$$g(\beta) = \lim \psi_\beta(k'_n, l'_n) = \lim \psi_\beta(k''_n, l''_n).$$

For each n, we can find a convex combination (k_n, l_n) of (k'_n, l'_n) and (k''_n, l''_n) with the property that

$$k_n = l_n - \frac{1}{2}.$$

By Lemma 5.4, $\psi_\beta(k_n, l_n)$ lies in an interval whose endpoints are $\psi_\beta(k'_n, l'_n)$ and $\psi_\beta(k''_n, l''_n)$, whence

$$g(\beta) = \lim \psi_\beta(k_n, l_n).$$

If we let $r_n = k_n + l_n$, we see that

$$\psi_\beta(k_n, l_n) = (2r_n + 1)/(6r_n + 7),$$

so that

$$g(\beta) = lim(2r_n + 1)/(6r_n + 7) \ge (2R + 1)/(6R + 7).$$

Bounds for $\zeta(5/7 + it)$. The technique of the last section can be adapted to bound $\zeta(\sigma + it)$ for any fixed σ in the range $1/2 < \sigma < 1$. To illustrate this, we will show that

$$\zeta(5/7 + it) \ll t^{(7R-3)/(14R+28)+\epsilon} \tag{5.4.2}$$

for every $\epsilon > 0$. For comparison purposes, note that

$$\frac{7R-3}{14R+28} = .07077534556548856\ldots$$

In Theorem 2.12, we obtained the exponent

$$\frac{1}{14} = .07142857142857143\ldots$$

and in Theorem 4.2 we obtained the exponent

$$\frac{368}{5159} = .07133165342120566\ldots.$$

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If (k,l) is an exponent pair and $N < N_1 \le 2N$ then

$$\sum_{N<n\le N_1} n^{-5/7-it} \ll t^{\theta_\alpha(k,l)},$$

where

$$\theta_\alpha(k,l) = \frac{(1-\alpha)k + \alpha l - 5\alpha}{7}.$$

Let

$$g(\alpha) = \inf \theta_\alpha(k,l),$$

and set $\delta = \sup\{g(\alpha) : 0 \le \alpha \le 1\}$. We will show that $\delta = (7R-3)/(14R+28)$. By Lemma 2.11, this is sufficient to prove (5.4.2).

Suppose first that $\alpha \le 1/3$, and let

$$(k,l) = A^2BA^2BAB(0,1) = (\frac{11}{186}, \frac{25}{31}).$$

Since $l - k - 5/7 > 0$, it follows that $\theta_\alpha(k,l)$ is an increasing function of α, and so

$$g(\alpha) \le \theta_{\frac{1}{3}}(k,l) = \frac{137}{1953} < \frac{7R-3}{14R+28}$$

for $\alpha \le 1/3$.

Next, suppose that $\alpha \ge 9/25$, and let

$$(k,l) = ABABA^2BAB(0,1) = (\frac{4}{49}, \frac{75}{98}).$$

Since $l - k + 5/7 < 0$, we see that

$$g(\alpha) \le \theta_{\frac{9}{25}}(k,l) = \frac{173}{2450} < \frac{7R-3}{14R+28}$$

for $\alpha \ge 9/25$.

Henceforth, we assume that $1/3 < \alpha < 9/25$. For α in this range, we have

$$\inf \theta_\alpha = \inf \theta_\alpha A = \frac{7k + (7l - 10k - 3)\alpha}{14k + 14}.$$

Let $\psi_\alpha(k,l) = \theta_\alpha A(k,l)$. We claim that for every $(k,l) \in A\mathbf{P}$, $\psi_\alpha(k,l)$ is an increasing function of α, and that for every $(k,l) \in BA\mathbf{P}$, $\psi_\alpha(k,l)$ is a decreasing function of α. For if $(k,l) = A(\kappa,\lambda)$, then

$$7l - 10k - 3 = \frac{7(\lambda - \kappa) + (1 - 2\kappa)}{2\kappa + 2} \ge 0, \tag{5.4.3}$$

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and if $(k,l) = BA(\kappa,\lambda)$ then

$$7l - 10k - 3 = \frac{16(\kappa-\lambda)+(2-4\lambda)}{2\kappa+2} \leq 0. \tag{5.4.4}$$

We will now show that

$$\inf \psi_\alpha \leq \frac{7R-3}{14R+28}$$

for all α under consideration. Let $(k_1,l_1) \in A\mathbf{P}$ and let $(k_2,l_2) \in BA\mathbf{P}$. We can find a convex combination (k,l) of (k_1,l_1) and (k_2,l_2) such that $7l-10k-3=0$. For this exponent pair,

$$\psi_\alpha(k,l) = \frac{k}{2k+2} = \frac{7(k+l)-3}{14(k+l)+28}.$$

Since

$$\psi_\alpha(k,l) \geq \min(\psi_\alpha(k_1,l_1), \psi_\alpha(k_2,l_2)),$$

it follows that

$$\inf \psi_\alpha \leq \inf \frac{7(k+l)-3}{14(k+l)+28} = \frac{7R-3}{14R+28}.$$

To complete the proof, we will now show that there is some β with $1/3 < \beta < 9/25$ and

$$\inf \psi_\beta \geq \frac{7R-3}{14R+28}. \tag{5.4.5}$$

Now let $g_A(\alpha) = \inf \psi_\alpha A$ and $g_B(\alpha) = \inf \psi_\alpha BA$, so that $\inf \psi_\alpha = \min(g_A(\alpha), g_B(\alpha))$. From the inequalities (5.4.3) and (5.4.4), we see that g_A is an increasing function of α and that g_B is a decreasing function of α. From Theorem 5.5, we see that $g_A(1/3) < g_B(1/3)$ and $g_B(9/25) < g_A(9/25)$. Since g_A and g_B are continuous, there is some β with $1/3 < \beta < 9/25$ and $g_A(\beta) = g_B(\beta)$.

Let (k_n', l_n') be a sequence of exponent pairs in $A\mathbf{P}$ such that

$$\inf \psi_\beta A = \lim_{n\to\infty} (k_n', l_n').$$

Let (k_n'', l_n'') be a sequence of exponent pairs in $BA\mathbf{P}$ such that

$$\inf \psi_\beta BA = \lim_{n\to\infty} (k_n'', l_n'').$$

Since $7l_n' - 10k_n' - 3 \geq 0$ and $7l_n'' - 10k_n'' - 3 \leq 0$, we can find a convex combination (k_n, l_n) of (k_n', l_n') and (k_n'', l_n'') such that $7l_n - 10k_n - 3 = 0$. We then have

$$\psi_\beta(k_n,l_n) = \frac{7((k_n+l_n)-3}{14((k_n+l_n)+28}.$$

Since $\psi_\beta(k_n,l_n) \leq \max(\psi_\beta(k_n',l_n'), \psi_\beta(k_n'',l_n''))$, statement (5.4.5) follows.

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5.5 NOTES

The material in this chapter is taken from Graham (1985), which was the first published account of computing optimal exponent pairs for general θ. Phillips and Rankin had earlier considered the special case $\theta(k,l) = k + l$. Phillips (1935) stated that he computed the first 12 terms of the q-sequence. He did not publish them; he stated only that he failed to find any pattern. Rankin (1955) gave the first eleven terms of the q-sequence for $\inf(k+l)$, and this gives the numerical value to 10 decimal places. He did the calculations during the 1940's on an adding machine. He did not publish the details of his method; the proofs, he said, "are rather long and involve much heavy algebra." He also stated that all the terms in the q-sequence of $\inf(k+l)$ satisfy $q \leq 3$. Heath-Brown (private communication) has established a corresponding result for more general θ: if θ satisfies (5.2.2), then every term in the q-sequence after the first is ≤ 3.

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